PROBLEM STATEMENT

The B2B Travel Crisis & Distribution Bottlenecks

Why current agent tools and generic LLM chatbots fail travel agents today.

1. Fragmentation

OPERATIONAL LAG

- Agents juggle 6-8 disconnected portals (GDS, hotel aggregators, sightseeing suppliers).
- Stitching flights + hotels + transfers takes 45-90 minutes per customized quote.
- High customer drop-off (>58%) due to quote latency.

2. Chatbot Failures

TECHNICAL GAP

- Generic AI chatbots hallucinate rates, expired flight seats, and hotel availability.
- Zero real-time price locking or inventory reservation capability.
- Unable to handle complex agency wallet ledgers, credit limits, or multi-city PNRs.

3. Support Overload

SERVICE PAIN

- 65%+ of operations workload spent on routine post-booking tickets (vouchers, date changes).
- Flight delays cause panic rebooking across hotel and transfer operators.
- High operational headcount costs and slow resolution times.

EXECUTIVE SOLUTION

VoyAgent AI: The Autonomous Travel Copilot Architecture

A purpose-built multi-agent ecosystem engineered directly for TBO's B2B travel platform.

1. Scoping Engine

PHASE 1: PLANNING

- Natural language intent parser extracts dates, pax, budget, and travel preferences.
- Intelligent follow-up questions clarify ambiguity before initiating searches.
- Builds persistent traveler context across multi-session inquiries.

2. Dynamic Bundler

PHASE 2: BOOKING

- Parallel querying across Flights (GDS/NDC), Hotel bedbanks, and Tours.
- Synthesizes day-by-day itineraries with optimal transit buffers and opening times.
- Real-time TBO API inventory locking with zero quote price slippage.

3. Autonomous Concierge

PHASE 3: SUPPORT

- Active Flight Sentinel: tracks airline delays and reschedules ground transfers.
- Self-service WhatsApp voucher & invoice retrieval for travelers.
- Human-in-the-Loop safeguards on all cancellations and wallet debits.